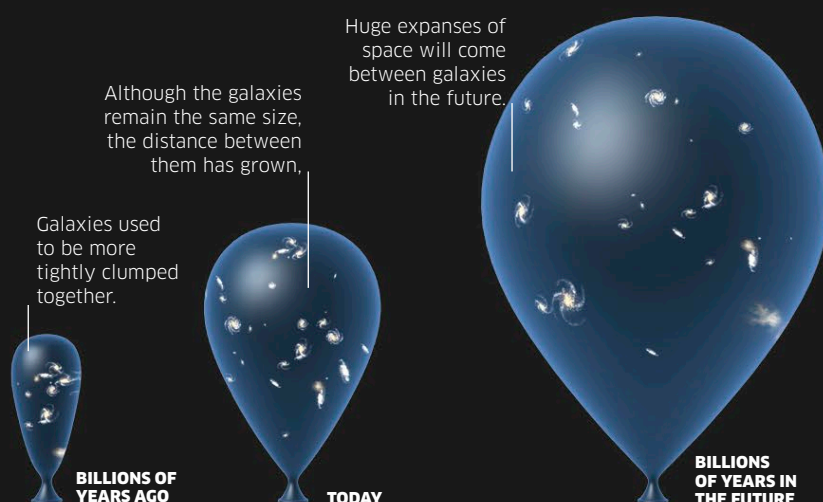


SPACE AND EARTH

All of space, matter, energy, and time make up the universe—a vast, ever-expanding creation that is so big it would take billions of years to cross it, even when traveling at the speed of light. Within the universe are clumps of matter called galaxies, and within those are planets like our own—Earth.

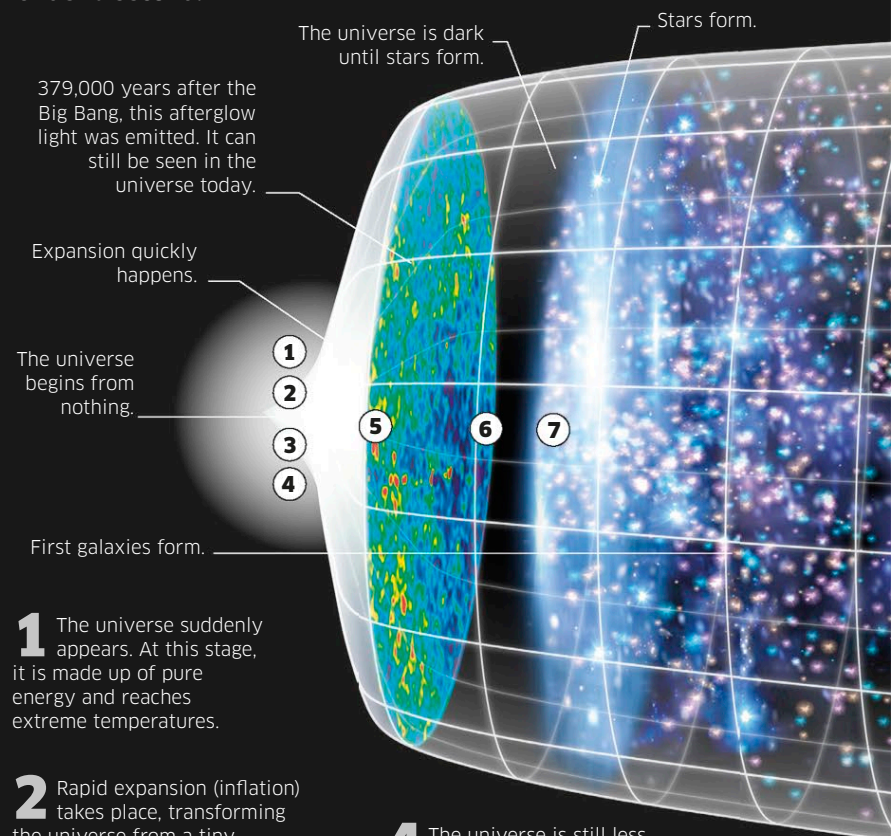
THE EXPANSION OF SPACE

Astronomers on Earth can observe galaxies moving away from us, but in reality they are moving away from every other point in the universe as well. These galaxies are not moving into new space—all of space is expanding and pulling them away from each other. This effect can be imagined by thinking of the universe as a balloon. As the balloon inflates, the rubber stretches and individual points on it all move further away from each other.



THE BIG BANG

The universe came into existence around 13.8 billion years ago in a cataclysmic explosion known as the Big Bang. Starting out as tinier than an atom, it rapidly expanded—forming stars, and clusters of stars called galaxies. A large part of this expansion happened incredibly quickly—it grew by a trillion kilometers in under a second.



1 The universe suddenly appears. At this stage, it is made up of pure energy and reaches extreme temperatures.

2 Rapid expansion (inflation) takes place, transforming the universe from a tiny mass smaller than a fraction of an atom into a gigantic space the size of a city.

3 Matter is created from the universe's energy. This starts out as minuscule particles and antiparticles (the same mass as particles but with an opposite electric charge). Many of these converge and cancel each other out, but some matter remains.

4 The universe is still less than a second old when the first recognizable subatomic particles start to form. These are protons and neutrons—the particles that make up the nucleus of an atom.

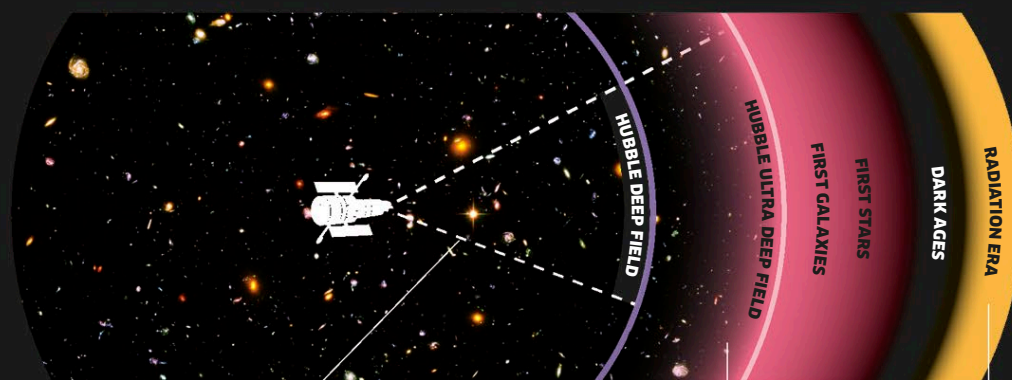
5 Over the next 379,000 years, the universe slowly cools, until eventually atoms are able to form. This development changes the universe from a dense fog into an empty space punctuated by clouds of hydrogen and helium gas. Light can now pass through it.

THE OBSERVABLE UNIVERSE

When we look at distant objects in the night sky, we are actually seeing what they looked like millions, or even billions, of years ago, because that is how long the light from them has taken to reach us. All of the space we can see from Earth is known as the observable universe. Other parts lie beyond that, but are too far away for the light from them to have reached us yet. However, using a space-based observatory such as the Hubble Space Telescope, we can capture images of deep space and use them to decipher the universe's past.

Hubble imaging

The Hubble Space Telescope has been operating since 1990 and has captured thousands of images of the universe. Many of these have been compiled to create amazing views of the furthest (and therefore oldest) parts of the universe we can see. These are known as Deep Field images.



The first Hubble Deep Field observed one part of the night sky over 10 days. It revealed galaxies formed less than a billion years after the Big Bang.

The later Hubble Ultra Deep Field image (above) shows even further into the past, picturing galaxies formed 13 billion years ago, when the Universe was between 400 and 700 million years old.

There are regions of space further back in time that Hubble and other powerful space telescopes cannot see.